

```
In [1]: import sys  
sys.version
```

```
Out[1]: '3.13.9 | packaged by Anaconda, Inc. | (main, Oct 21 2025, 19:09:58) [MSC v.192  
9 64 bit (AMD64)]'
```

```
In [2]: import keyword  
keyword.kwlist
```

```
Out[2]: ['False',  
         'None',  
         'True',  
         'and',  
         'as',  
         'assert',  
         'async',  
         'await',  
         'break',  
         'class',  
         'continue',  
         'def',  
         'del',  
         'elif',  
         'else',  
         'except',  
         'finally',  
         'for',  
         'from',  
         'global',  
         'if',  
         'import',  
         'in',  
         'is',  
         'lambda',  
         'nonlocal',  
         'not',  
         'or',  
         'pass',  
         'raise',  
         'return',  
         'try',  
         'while',  
         'with',  
         'yield']
```

```
In [3]: len(keyword.kwlist)
```

```
Out[3]: 35
```

Identifiers

```
In [4]: lvar = 10
```

```
Cell In[4], line 1
  lvar = 10
    ^
SyntaxError: invalid decimal literal
```

```
In [5]: val2@ = 35
```

```
Cell In[5], line 1
  val2@ = 35
    ^
SyntaxError: invalid syntax
```

```
In [6]: import = 125
```

```
Cell In[6], line 1
  import = 125
    ^
SyntaxError: invalid syntax
```

```
In [7]: # identifier can be combination of upper, lower case letter and also _ underscore
# it cannot start with num but we can use num after any letter
# we cannot use special character
```

```
In [8]: val1 = 10
val1
```

```
Out[8]: 10
```

```
In [9]: p = 20
q = 20
r = q
p, type(p), hex(id(p))
```

```
Out[9]: (20, int, '0x7ffc4ed7a608')
```

```
In [10]: q, type(q), hex(id(q))
```

```
Out[10]: (20, int, '0x7ffc4ed7a608')
```

```
In [11]: r, type(r), hex(id(r))
```

```
Out[11]: (20, int, '0x7ffc4ed7a608')
```

```
In [12]: p = 20
p = p + 10
p
```

```
Out[12]: 30
```

```
In [13]: intvar = 10
floatvar = 2.57
strvar = 'python language'
print(intvar)
print(floatvar)
print(strvar)
```

```
10
2.57
python language
```

```
In [14]: # multiple assignment

intvar , floatvar , strvar = 10,2.57,'python'
print((intvar,floatvar,strvar))

(10, 2.57, 'python')
```

Data Types

```
In [15]: v1 = 10
print(v1)
print(type(v1))
print(sys.getsizeof(v1))
print(v1, 'is integer?', isinstance(v1,int))

10
<class 'int'>
28
10 is integer? True
```

```
In [17]: v2 = 20.47
print(v2)
print(sys.getsizeof(v2))
print(v2, 'is float?', isinstance(v2,float))

20.47
24
20.47 is float? True
```

```
In [18]: v3 = 20 + 3j
print(v3)
print(sys.getsizeof(v3))
print(v3, 'is complex?', isinstance(v3,complex))

(20+3j)
32
(20+3j) is complex? True
```

```
In [19]: # Boolean
```

```
In [20]: b1 = True
```

```
In [21]: b2 = False
```

```
In [22]: print(type(b1))
```

```
<class 'bool'>
```

```
In [23]: bool(0)
```

```
Out[23]: False
```

```
In [24]: bool(1)
```

Out[24]: True

In [25]: `bool()`

Out[25]: False

In [26]: *# Strings*

In [27]: `s1 = 'hello python'`
`print(s1)`

hello python

In [29]: `s2 = "hello world"`
`s2`

Out[29]: 'hello world'

In [30]: `s3 = '''Hello`
`world'''`
`print(s3)`

Hello
world

In [32]: `s4 = ('Happy '`
`'Monday '`
`'Everyone')`
`s4`

Out[32]: 'Happy Monday Everyone'

In [35]: `s5 = 'whoom '`
`s5 = s5 * 5`
`s5`

Out[35]: 'whoom whoom whoom whoom whoom '

In [36]: `len(s5)`

Out[36]: 30

In [37]: *# string indexing*

In [38]: `s1`

Out[38]: 'hello python'

In [39]: `s1[0]`

Out[39]: 'h'

In [40]: `s1[5]`

Out[40]: ' '

In [41]: `s1[6]`

Out[41]: 'p'

In [42]: `s1[-1]`

Out[42]: 'n'

In [44]: `# string slicing`
`s1[0:5]`

Out[44]: 'hello'

In [45]: `s1[6:12]`

Out[45]: 'python'

In [46]: `s1[-6:]`

Out[46]: 'python'

In [47]: `del s4`

In [48]: `# string concatenation`

In [51]: `s1 = 'hello'`
`s2 = 'nit'`
`s3 = s1 + s2`
`print(s3)`

hellonit

In [52]: `# complex`

In [53]: `z = 20 + 3j`
`w = 10 + 2j`
`z.imag`

Out[53]: 3.0

In [54]: `z.real`

Out[54]: 20.0

In [55]: `print(z + w)`

(30+5j)

In [56]: `print(z - w)`

(10+1j)

In [57]: `print(z * w)`

(194+70j)

In [58]: `print(z / w)`

(1.9807692307692308-0.09615384615384615j)

In []: